

Curriculum Vitae

[William Troiani](#)

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Education:

2020 - 2024 **University of Melbourne, studied abroad at the University of Paris Thirteen (Université Sorbonne Paris Nord):** Ph.D. in Mathematics, 2024.

Thesis Title: [Algebraic Geometry and Linear Logic](#)

Supervisors: Daniel Murfet, Thomas Seiller (Université Sorbonne Paris Nord).

2017 - 2019 **University of Melbourne:** Master of Science (Mathematics and Statistics).

Thesis Topic: [Simplicial Sets are Algorithms](#)

Supervisor: Daniel Murfet.

2013 - 2016 **University of Melbourne:** Bachelor of Science (Mathematics and Statistics).

Employment:

2024 – present **Postgraduate Researcher (ARC-funded):** University of Melbourne

Supervisors: Daniel Murfet, David Ridout.

2017 – 2018 **Research Assistant:** Funded by Defence Science and Technology (DST)

Supervisor: Daniel Murfet.

Interests: Mathematical logic, computer science, algebraic geometry.

Publications:

[The Internal Logic and Finite Colimits](#) (Solo-authored) Logica Universalis (published).

[Elimination and Cut-elimination in Multiplicative Linear Logic](#) (Co-authored with Daniel Murfet) Mathematical Structures in Computer Science (accepted).

Preprints (all submitted to Mathematical Structures in Computer Science, all co-authored with Daniel Murfet):

[Gentzen-Mints-Zucker Duality](#)

[Linear Logic and Quantum Error Correcting Codes](#)

[Linear Logic and the Hilbert Scheme](#)

[Programs as Singularities](#)

Lecture series:

September 2022: Student seminar on category theory (undergraduate course at Université Sorbonne Paris Nord).

Supervision:

September 2023: Undergraduate research project at the Université Sorbonne Paris Nord.
Thesis title: [Le théorème de Nielsen-Schreier, deux approches différentes](#) (The Nielsen–Schreier Theorem: Two Different Approaches).

2025: Mentor of the Noisy Turing Machine Research Group (PhD students) at the University of Melbourne.

Contact: Daniel Murfet.

Events organised:

November 2023: [Derivatives in Logic and Learning bonus seminar on Differential Linear Logic.](#)

Scholarships:

2015: **University of Melbourne:** Vacation Research Scholarship.
Supervisor: Peter Forrester.

2016: **AMSI Summer Research Scholarship:** Vacation Research Scholarship.
Supervisor: Daniel Murfet.

2016: **AMSI Travel Grant.**

Other academic activity:

2022: Informally lectured two courses in logic. One on Gödel’s First Incompleteness Theorem, and one on The Ax-Grothendieck Theorem (including basic model theory) at [Metauni](#).

2022: Informally gave a free course on category theory for computer science titled *Shadows of Computation* at [Metauni](#).

2022 & 2025: Have presented several lectures on the connections between mathematics and music, for example at the [ART Seminar \(University of Melbourne\)](#).

2021: Collaborated with an artist on a piece titled [Taming the Frame](#).

Academic References:

[Daniel Murfet.](#)
[Marcy Robertson.](#)
[Morgan Rogers.](#)
[Thomas Seiller.](#)